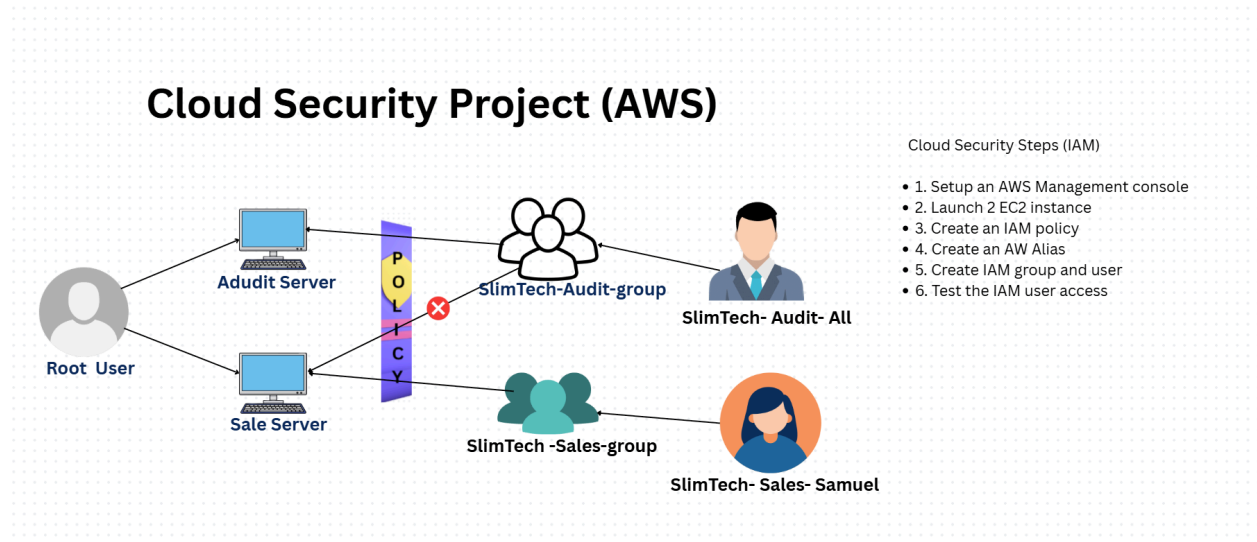


AWS IAM Cloud Security Project

1. Project Overview

I completed this project on cloud security controls in Amazon Web Services (AWS), focusing on Identity and Access Management (IAM). The goal was to create a least-privilege policy, attach it to a user group, and verify that the policy correctly restricts actions on two Amazon EC2 instances (audit and sales).



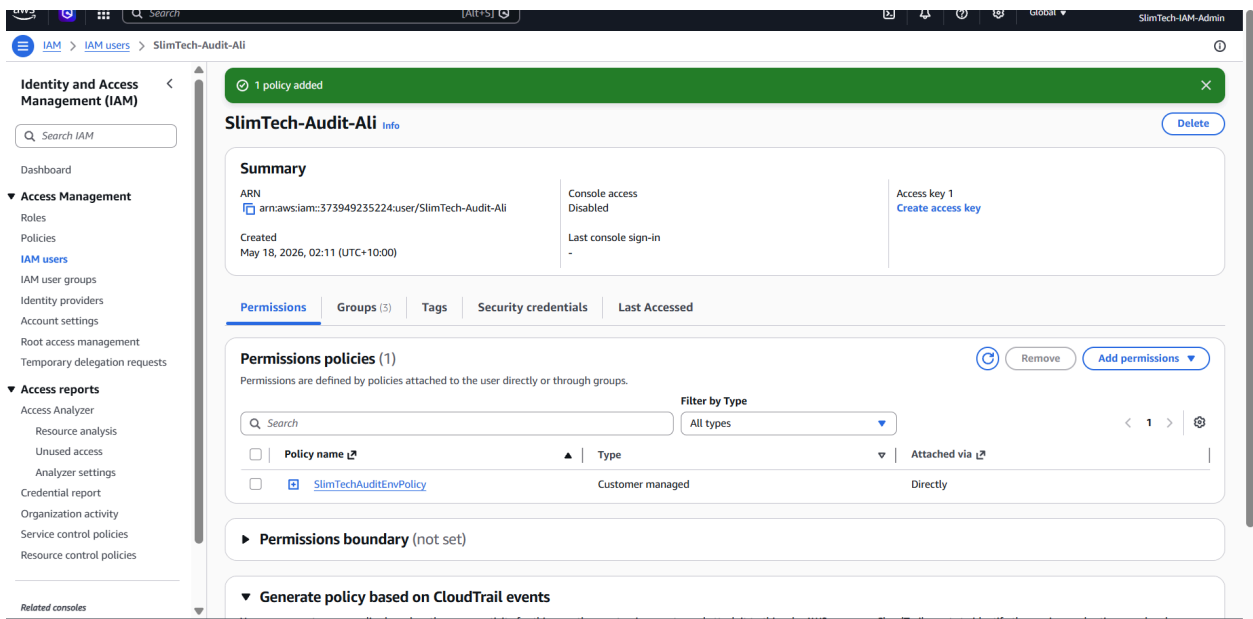
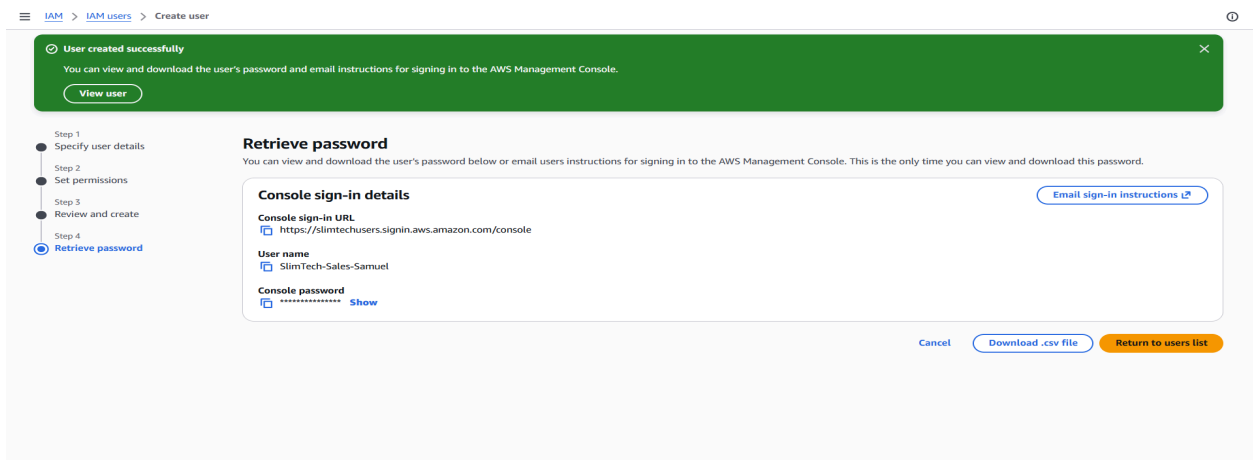
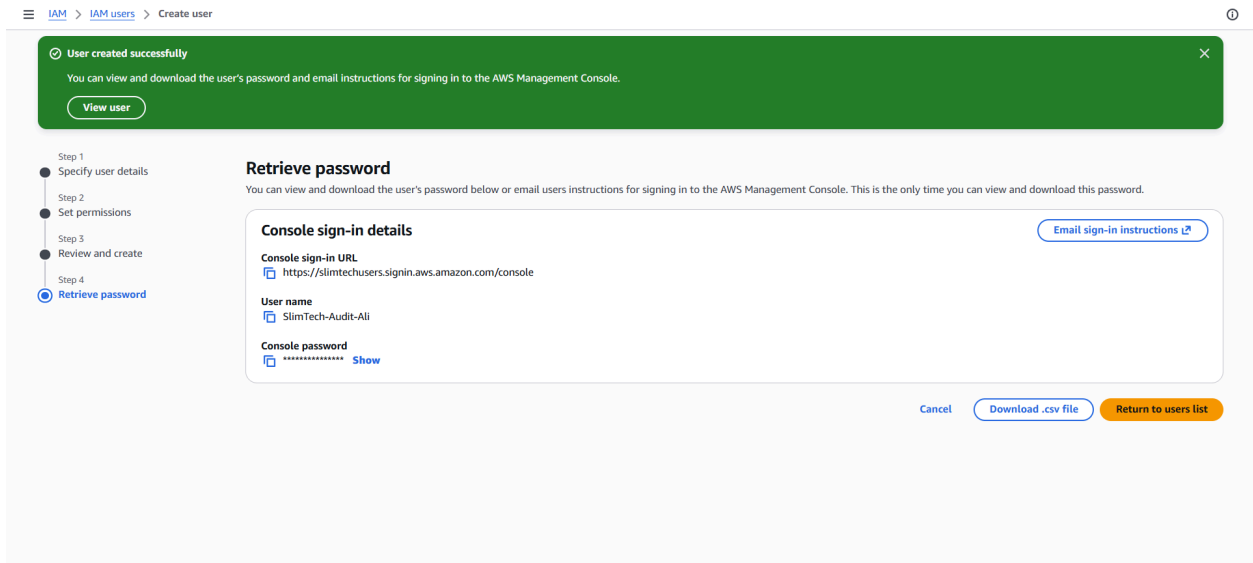
Tools & Concepts

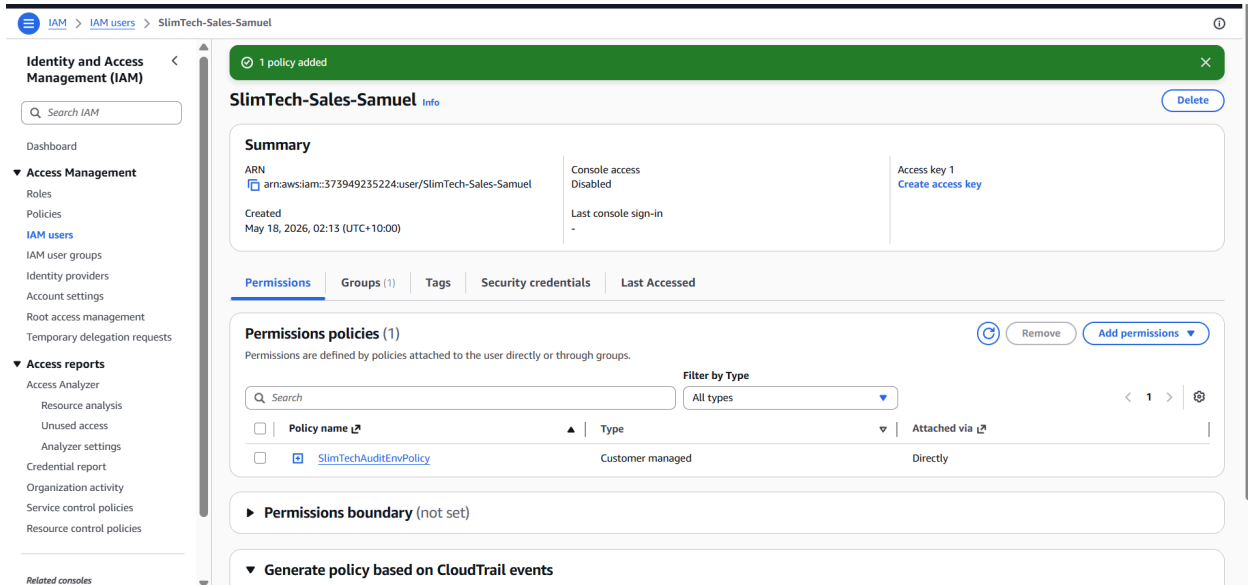
- AWS IAM – users, groups, policies, account alias
- Amazon EC2 – instance tagging and lifecycle actions
- JSON policy syntax – Effect, Action, Resource
- Principle of least privilege and policy testing

Tagging Strategy

Instance	Tag Key	Tag Value
SlimTech-Audit-Ali	Environment	Audit
SlimTech-Sales-Samuel	Environment	Sales

Tags were used to control policy logic and enforce targeted permissions.

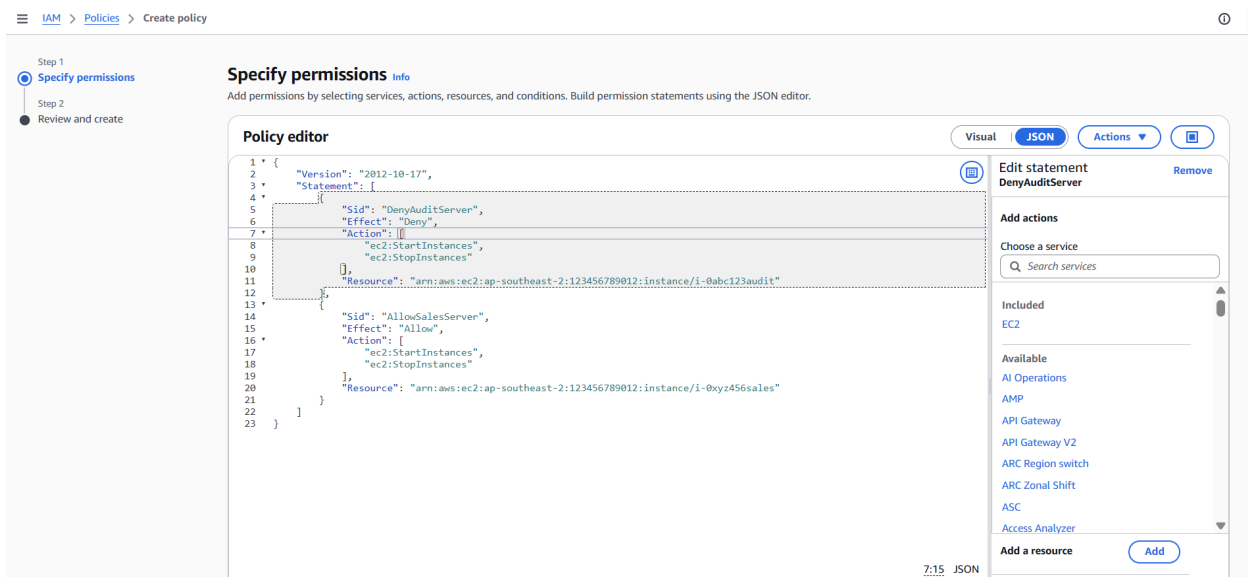




AM Policy Implementation

A custom JSON IAM policy was created to:

- Deny start/stop actions on the SlimTech-Audit-Server instance
- Allow start/stop actions on the SlimTech -Sales-Server instance



Account Alias

I set a memorable account alias to replace the default numeric URL, making sign-in easier for team members.

AWS Account

Account ID
373949235224

Account Alias
slimtechusers [Edit](#) | [Delete](#)

Sign-in URL for IAM users in this account
<https://slimtechusers.signin.aws.amazon.com/console>

aws

Search

[Alt+S]

Asia Pacific (Sydney)

SlimTech-IAM-Admin

Console home

Reset to default layout

Add widgets

Recently visited

IAM

EC2

AWS Global View

Billing and Cost Management

S3

Amazon EventBridge

Simple Notification Service

CloudTrail

View all services

Applications (0)

Create application

Region: Asia Pacific (Sydney)

Select Region
ap-southeast-2 (Current Region)

Find applications

< 1 >

Name

Description

Region

Orig

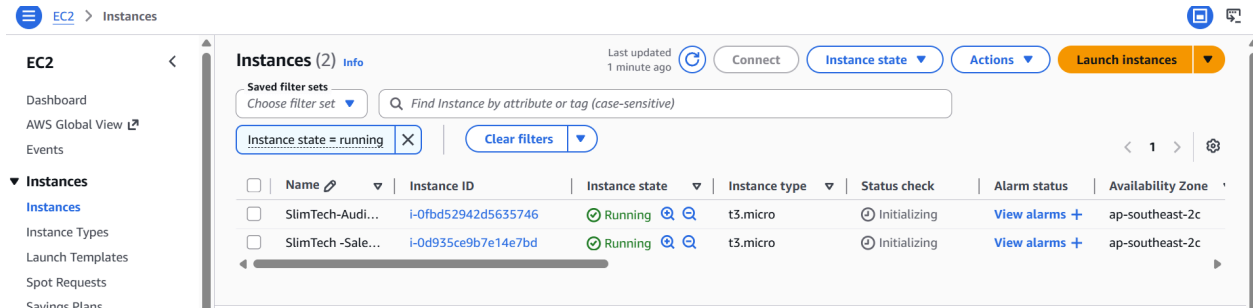
No applications
Get started by creating an application.

Create application

Go to myApplications

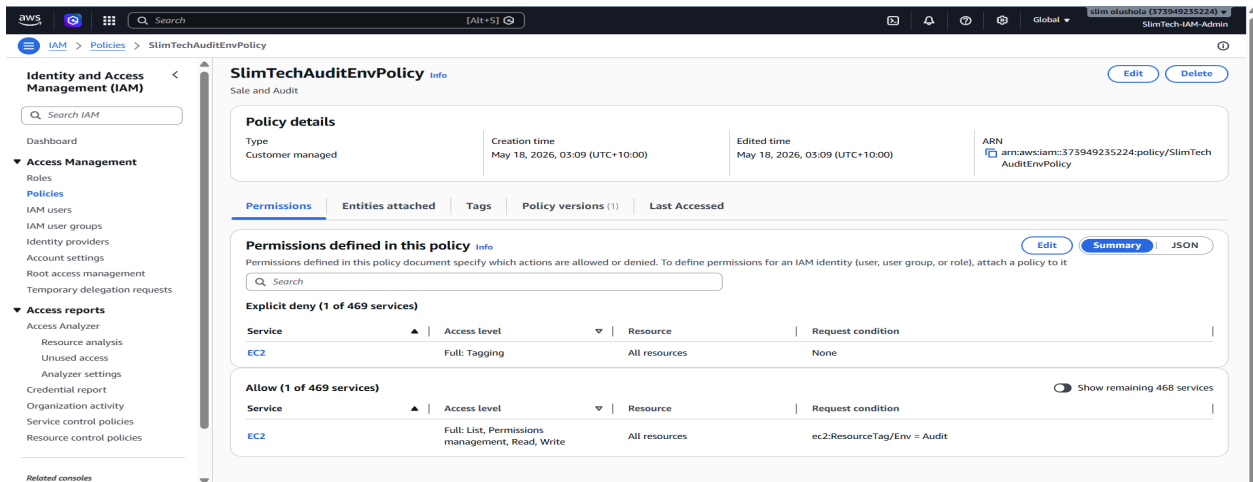
IAM Users & Groups

1. Created an IAM user group called Developers.
2. Attached the **SlimTechAuditEnvPolicy** policy to the group.
3. Added individual IAM users who require controlled EC2 access



The screenshot shows the AWS Management Console for the EC2 service. The left sidebar contains navigation links for EC2, Dashboard, AWS Global View, Events, Instances, Instance Types, Launch Templates, Spot Requests, and Savings Plans. The main content area is titled 'Instances (2)' and includes a search bar, filter buttons, and a table of instances.

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
☐	SlimTech-Audi...	i-0fbd52942d5635746	Running	t3.micro	Initializing	View alarms +	ap-southeast-2c
☐	SlimTech -Sale...	i-0d935ce9b7e14e7bd	Running	t3.micro	Initializing	View alarms +	ap-southeast-2c



The screenshot shows the AWS Management Console for the IAM service. The left sidebar contains navigation links for Identity and Access Management (IAM), Dashboard, Access Management, Roles, Policies, IAM users, IAM user groups, Identity providers, Account settings, Root access management, Temporary delegation requests, Access reports, Access Analyzer, Resource analysis, Unused access, Analyzer settings, Credential report, Organization activity, Service control policies, and Resource control policies. The main content area is titled 'SlimTechAuditEnvPolicy' and includes a search bar, filter buttons, and a table of permissions.

Service	Access level	Resource	Request condition
EC2	Full: Tagging	All resources	None

Explicit deny (1 of 469 services)

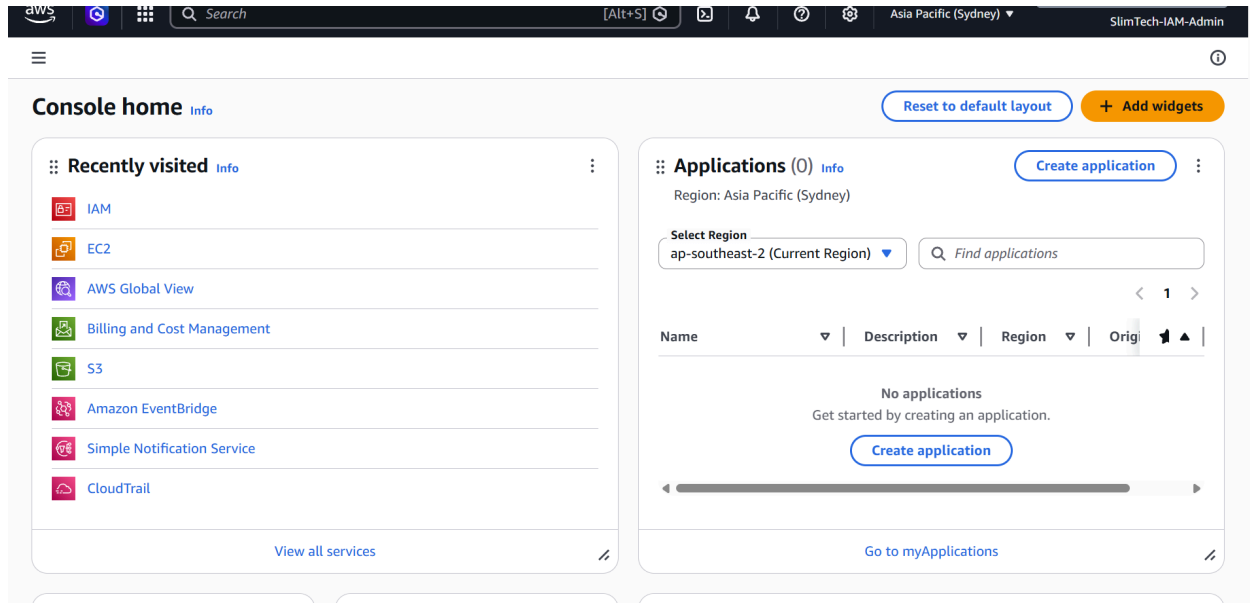
Service	Access level	Resource	Request condition
EC2	Full: List, Permissions management, Read, Write	All resources	ec2:ResourceTag/Env = Audit

Allow (1 of 469 services)

Logging in as an IAM User

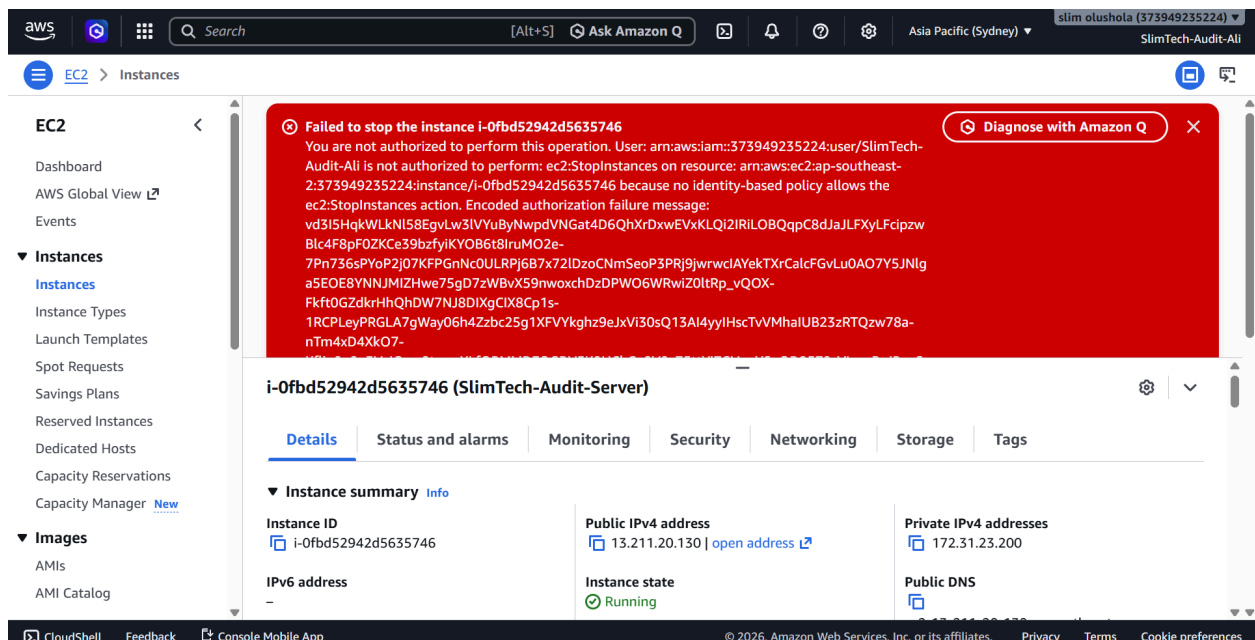
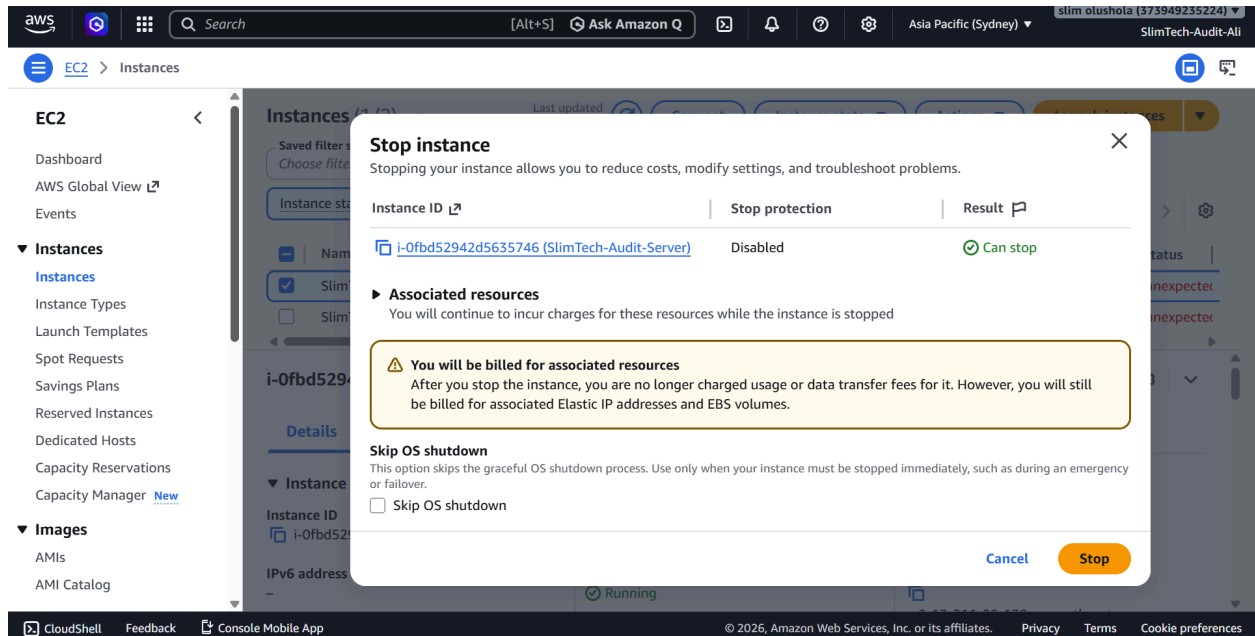
IAM users can sign in through:

- AWS Management Console (using the new alias URL)
- AWS CLI via programmatic keys



Access Testing Results

Action	Expected	Result
Stop SlimTech-Audit-Server Instance	Denied	Access denied
Start SlimTech-Audit-Server Instance	Denied	Access denied
Stop SlimTech -Sales-Server Instance	Allowed	Success
Start SlimTech -Sales-Server Instance	Allowed	Success



Security Concepts Demonstrated

- Principle of Least Privilege
- Role-Based Access Control
- Cloud Logging & Monitoring
- Identity Federation Preparation
- Resource-Level Permissions
- Security Validation Testing

Project Workflow Summary

1. Created cloud account and secured root access with MFA
2. Created admin IAM user and enabled MFA
3. Created IAM group and users
4. Built EC2 infrastructure
5. Applied tagging structure
6. Created custom IAM policy
7. Configured CloudTrail, SNS, and EventBridge
8. Tested user access permissions and restrictions

Key Skills Demonstrated

- Cloud Security Implementation
- IAM Policy Design
- JSON Policy Writing
- Access Control Testing
- AWS Monitoring Tools
- Security Validation Procedures

Purpose

This lab simulates real-world cloud security engineering tasks and demonstrates practical knowledge of designing, enforcing, and validating access control policies in enterprise cloud environments.

Author

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